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ABSTRACT

Since dynamic mode decomposition (DMD) Galerkin reduced-order models (ROMs) have been developed and successfully applied across various fields, they hold great potential for constructing parametric reduced-order models (PROMs). By integrating the DMD-Galerkin framework with different interpolation methods, accurate and efficient DMD-Galerkin PROMs are effectively constructed. Specifically, two interpolation methods—Grassmann manifold interpolation (GMI) and mode-realigned pointwise interpolation (MRPWI)—are employed to obtain complex-valued, non-orthonormal DMD modes at target (unseen) parameters. The interpolated DMD modes are then incorporated into the Galerkin projection to build the DMD-Galerkin PROMs. It is noteworthy that the GMI, rigorously derived based on the complex Grassmann manifold via the noncompact Stiefel manifold representation, and the MRPWI, which relies on Kasner's pseudo-angle, are both directly applied to interpolate complex-valued, non-orthonormal DMD modes, making them particularly suitable for this interpolation process. To evaluate the effectiveness of the proposed DMD-Galerkin PROMs, flow over a cylinder is adopted as a canonical benchmark for assessing their predictive fidelity. The results show that for target parameters within the parameter space, the DMD-Galerkin PROMs yield sufficiently accurate predictions compared to DNS and DMD-Galerkin ROMs, while providing computational efficiency over high-fidelity simulations. Furthermore, the DMD-Galerkin PROMs constructed with GMI and MRPWI exhibit comparable accuracy, while those with the latter are more efficient due to its theoretical simplicity.

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I. INTRODUCTION

Large-scale dynamical systems, particularly in the field of fluid mechanics, often depend on a set of parameters. Investigating the complex underlying physical phenomena in parameter-dependent systems typically demands prohibitively high computational resources to capture sufficient detail with high accuracy. This challenge has driven the development of low-dimensional parametric reduced-order models (PROMs), which aim to efficiently and accurately approximate full-order dynamical systems while achieving significant dimensionality reduction.¹

Parametric reduced-order modeling can be broadly classified into intrusive (projection-based) and nonintrusive (data-driven) categories,² both of which can be conducted through either a local approach (interpolation-based) or a global approach (stacking-based).³ Additionally, neural networks (NNs) have emerged as a promising approach for developing data-driven reduced-order models (ROMs),⁴

though their application to parameterized systems remains limited.⁵ Data-driven PROMs, owing to their low implementation complexity and independence from prior knowledge of governing equations, have garnered significant attention and widespread application in recent years. Nevertheless, the absence of first-principles constraints often compromises their accuracy, stability, and generalization.^{2,6} In contrast, projection-based methods—which utilize the underlying governing equations—are generally more structured and theoretically well-founded.³ Specifically, projection-based approaches are constructed by projecting the governing equations onto a low-dimensional subspace through Galerkin projection, where the subspace is spanned by a modal basis derived from dimensionality reduction techniques.

Regarding dimensionality reduction techniques, several studies provide systematic reviews, including proper orthogonal decomposition (POD),^{7,8} dynamic mode decomposition (DMD)⁹ and others.^{1,10,11} DMD, first developed by Schmid⁹ Schmid and Sesterhenn¹² and later

elaborated by Rowley *et al.*,¹³ is a powerful and rapidly evolving dimensionality reduction technique for identifying dynamical systems from high-dimensional data.^{14,15} It is widely recognized for its straightforward numerical implementation and theoretical connection to underlying nonlinear dynamics via Koopman theory.^{14,15} In combination with the DMD method, many works have developed PROMs. To the best of our knowledge, the parametric DMD proposed in Ref. 16 is the first stacking-based DMD PROMs, and it should be noted that this approach also integrates pointwise Lagrange interpolation for unseen parameters. Building on this, for the local approach, Huhn *et al.*¹⁷ employed pointwise Lagrange interpolation to approximate either the low-dimensional linear operator or its eigenvalues and eigenvectors, thereby constructing DMD PROMs. Subsequently, manifold interpolation is utilized, based on the concept of the general linear group and the Grassmann manifold, to evaluate the low-dimensional linear operator and POD modes, respectively, for developing DMD PROMs.¹⁸ This approach preserves the intrinsic properties of these quantities involved in the interpolation process, ensuring that the resulting PROMs accurately capture the essential dynamics of the system and provide reliable approximations for unseen parameters.^{3,19} Furthermore, several works employing machine learning techniques, such as radial basis functions, k -nearest-neighbors regression, and weighted interpolated nearest-neighbors algorithms, have been integrated into the DMD method to construct PROMs for parameterized systems, yielding significant performance improvements in their test cases.^{20–22} These aforementioned strategies are widely investigated and applied, but they fall under the category of data-driven DMD PROMs.

Compared with data-driven approaches, projection-based methods utilize the fundamental governing equations and are potentially more flexible. Given DMD modes are non-orthonormal, Tissot *et al.*²³ and Callahan *et al.*²⁴ proposed DMD-Galerkin ROMs through oblique projection (Petrov-Galerkin method) instead of the standard Galerkin projection. Tissot *et al.*²³ further noted that the adjoint DMD modes suffer from numerical corruption. Subsequently, Zhang and Wei²⁵ developed DMD-Galerkin ROMs by leveraging the bi-orthogonality between the original DMD modes and their adjoint counterparts, successfully applying the approach to flow over a cylinder and capturing its essential dynamics. However, these studies focus on conventional ROMs, which require data from available cases, rather than PROMs.

In short, existing strategies that employ DMD can be broadly categorized as either data-driven PROMs or projection-based conventional ROMs. Furthermore, interpolation-based methods demonstrate superior suitability for constructing PROMs compared to stacking methods, which are fundamentally limited by computational cost issues, frequency inflexibility, and temporal sampling constraints.³ To date, both Grassmann manifold interpolation (GMI) and mode-realigned pointwise interpolation (MRPWI) have demonstrated superior performance in constructing data-driven DMD PROMs.²⁶ It is therefore necessary to extend these approaches to projection-based DMD PROMs.^{27,28} To accommodate complex-valued, non-orthonormal DMD modes, Du *et al.*²⁶ extended GMI to the complex Grassmann manifold based on the compact Stiefel manifold representation. However, this approach still requires orthonormalization preprocessing for interpolation. To address this limitation, the formulation of GMI on the complex Grassmann manifold via the noncompact Stiefel manifold representation is rigorously derived. Furthermore, the

MRPWI, proposed in Ref. 26, is noted for its accuracy, efficiency, and simplicity. These methods are incorporated here into the DMD-Galerkin framework for constructing DMD-Galerkin PROMs. Subsequently, a canonical problem—flow over a cylinder—is used to demonstrate the capabilities of the proposed PROMs, with specific attention to their predictive accuracy in sensitivity analysis and generalization performance.

The structure of this work is as follows: Sec. II outlines the construction of DMD-Galerkin PROMs, including the Galerkin projection, the DMD algorithm, and interpolation methods—GMI and MRPWI. Subsequently, Sec. III introduces the flow over a cylinder and the corresponding numerical framework. The evolution of flow fields and aerodynamic coefficients for the flow over a cylinder is then carefully examined to evaluate these PROMs. Finally, the conclusion is presented in Sec. IV. Additionally, Appendix provides the derivation of a theoretical framework for GMI, formulated on the complex Grassmann manifold via the noncompact Stiefel manifold representation.

II. DMD-GALERKIN PARAMETRIC REDUCED-ORDER MODELS

The methodology for developing DMD-Galerkin PROMs consists of three key components: (A) Galerkin projection for temporal evolution equations, (B) the DMD algorithm to obtain reduced-order bases, which together form the DMD-Galerkin framework, and (C) interpolation methods for basis adaptation at target (unseen) parameters.

A. DMD-Galerkin framework

The Galerkin projection transforms the infinite-dimensional partial differential equations into a finite set of ordinary differential equations (ODEs).²⁹ Here, the DMD-Galerkin framework are first introduced, with the governing equations projected onto a finite-dimensional subspace spanned by the modal basis obtained from the DMD method.

In the case of incompressible flow, the governing equations are given by the Navier-Stokes equations. The velocity and pressure are approximated using a truncated linear combination of the basis,^{30,31}

$$\begin{pmatrix} \mathbf{u}(\mathbf{x}, t) \\ p(\mathbf{x}, t) \end{pmatrix} \approx \sum_{l=1}^{N_r} \alpha_l(t) \begin{pmatrix} \boldsymbol{\varphi}_l(\mathbf{x}) \\ \boldsymbol{\psi}_l(\mathbf{x}) \end{pmatrix} = \sum_{l=1}^{N_r} \alpha_l(t) \boldsymbol{\phi}_l(\mathbf{x}), \quad (1)$$

where $\boldsymbol{\varphi}_l$ and $\boldsymbol{\psi}_l$ are the spatial modes for velocity and pressure, respectively, and $\alpha_l(t)$ are the corresponding temporal coefficients. Additionally, including pressure in the spatial modes not only enhances the stability of the ROMs but also facilitates the evaluation of aerodynamic coefficients.^{30,32}

Since the DMD modes are not orthonormal, the Petrov-Galerkin projection is employed to derive the DMD-Galerkin framework. A low-dimensional dynamical system is derived by applying the Galerkin projection to the Navier-Stokes momentum equations, incorporating the approximation from Eq. (1),²³

$$(\boldsymbol{\varphi}_l, \mathbf{u}_t) = (\boldsymbol{\varphi}_l, -\mathbf{u} \cdot \nabla \mathbf{u}) + (\boldsymbol{\varphi}_l, \nu \nabla^2 \mathbf{u} - \nabla p). \quad (2)$$

This leads to the evolution equation for the temporal coefficients, formulated as a system of N_r nonlinear ODEs,²⁴

$$M_{lm}\dot{\alpha}_l = \sum_{m=1}^{N_r} \sum_{n=1}^{N_r} Q_{lmn} \alpha_m \alpha_n + \sum_{m=1}^{N_r} L_{lm} \alpha_m. \quad (3)$$

Here, $(\cdot)$ denotes the time derivative. The mass M_{lm} , quadratic Q_{lmn} , and linear L_{lm} terms are given by,

$$\begin{aligned} M_{lm} &= (\varphi_l, \varphi_m), \quad Q_{lmn} = (\varphi_l, -\varphi_m \cdot \nabla \varphi_n), \\ L_{lm} &= (\varphi_l, \nu \nabla^2 \varphi_m - \nabla \psi_m). \end{aligned} \quad (4)$$

B. Dynamic mode decomposition

As mentioned previously, the DMD modes serve as the modal basis for the DMD-Galerkin framework, and thus the DMD method is briefly outlined. It is assumed that the data collected from the dynamical system are sampled at equally spaced time intervals. At each time t_j , the sampled data form a snapshot of the system state, which is then reshaped into a high-dimensional column vector $\mathbf{s}_j$. These snapshots are then assembled into the data matrices X and X' as follows:

$$\begin{aligned} X &= [\mathbf{s}_0, \mathbf{s}_1, \dots, \mathbf{s}_{N_s-1}] \in \mathbb{C}^{N_n \times N_s}, \\ X' &= [\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_{N_s}] \in \mathbb{C}^{N_n \times N_s}. \end{aligned} \quad (5)$$

In fluid dynamics, N_s is typically much smaller than the spatial dimension N_n . Specifically, for incompressible flow, each snapshot includes both the velocity and pressure fields, and is given by

$$\mathbf{s}_j = \begin{pmatrix} \mathbf{u}(\mathbf{x}, t_j) \\ p(\mathbf{x}, t_j) \end{pmatrix} \in \mathbb{R}^{N_n}. \quad (6)$$

The underlying dynamical system is approximated by relating the arranged data matrices, as described in Ref. 14,

$$X' \approx AX, \quad (7)$$

where A serves as a finite-dimensional approximation of the Koopman operator.¹⁵ To avoid explicitly forming the high-dimensional operator A , the DMD method employs a low-dimensional operator $\tilde{A}$ that approximates A .³³

The reduced operator $\tilde{A}$ is obtained by projecting A onto the truncated POD modes of X as follows:

$$\tilde{A} = U^* A U = U^* X' V \Sigma^{-1} \in \mathbb{C}^{N_r \times N_r}, \quad (8)$$

where $*$ denotes the conjugate transpose. Specifically, the reduced singular value decomposition (SVD) of X with a rank- N_r approximation is performed,

$$X \approx U \Sigma V^*. \quad (9)$$

Here, $U \in \mathbb{C}^{N_n \times N_r}$ represents the left singular vectors (POD modes), $V \in \mathbb{C}^{N_s \times N_r}$ denotes the right singular vectors, and the diagonal matrix $\Sigma \in \mathbb{C}^{N_r \times N_r}$ contains the non-negative singular values. Then, the eigendecomposition of $\tilde{A}$ is performed as³⁴

$$\tilde{A} W = W \Lambda, \quad (10)$$

where the columns of $W \in \mathbb{C}^{N_r \times N_r}$ represent the eigenvectors, and the corresponding eigenvalues are given by the diagonal elements of Λ . Importantly, since A and $\tilde{A}$ are similar matrices, they share the same nonzero eigenvalues. The eigenvectors of A , that is, the DMD modes,

can thus be recovered from the eigenvectors of $\tilde{A}$. Finally, the projected DMD modes are computed as the columns ϕ_j of Φ ,

$$\Phi = U W \in \mathbb{C}^{N_n \times N_r}, \quad (11)$$

which are employed in the DMD-Galerkin framework.

C. Interpolation methods

Interpolation methods, including Grassmann manifold interpolation (GMI) and mode-realigned pointwise interpolation (MRPWI), are introduced to construct DMD-Galerkin PROMs. Specifically, the DMD modes $\{\Phi_j\}_{j=1}^{N_p}$, obtained from the DMD method for the set of parameters $\{\gamma_j\}_{j=1}^{N_p}$, are interpolated to yield their counterparts at the target parameter $\bar{\gamma}$, which are then combined with Eq. (3). The accuracy of the GMI and MRPWI methods—that is, the performance of PROMs—is influenced by the number of modes N_r (i.e., truncation), the system parameter interval $\Delta\gamma$, and the number of neighbors N_p . It should be noted that the mode ranking criterion, as well as the rule for selecting both the N_p interpolating cases and the reference case, and the distance metrics developed in Ref. 26, are directly adopted in the interpolation process. The detailed description and demonstration can be found in Ref. 26.

1. Grassmann manifold interpolation

GMI offers a powerful and geometrically consistent framework, enabling effective subspace interpolation. Although the geodesic, exponential mapping, and logarithmic mapping on the complex Grassmann manifold have been presented in Ref. 26, orthonormalization is still required due to the reliance on the Stiefel manifold representation. Consequently, there remains potential for further development of interpolation methods for non-orthonormal DMD modes on the complex Grassmann manifold. To this end, detailed derivations of the geodesic, exponential mapping, and logarithmic mapping on the complex Grassmann manifold, formulated through the noncompact Stiefel manifold representation, are provided in Appendix.

GMI is performed by projecting a set of points (which are subspaces) that lie on the Grassmann manifold onto the tangent space defined at a reference point chosen from these subspaces, interpolating the projections, and then mapping the interpolated projection back to the manifold to obtain the desired point. Geometrically, a geodesic on the manifold is the locally shortest path originating from the reference point and ending at another sufficiently close neighboring point. The reference point serves as the initial position, while the projections of the neighboring points on the tangent space represent the initial derivatives, which uniquely determine their corresponding geodesics. In other words, all these geodesics share the same initial position but differ in their initial derivatives. These initial derivatives, associated with different parameters, are interpolated to obtain an initial derivative corresponding to the target parameter. This interpolated initial derivative defines a geodesic on the manifold whose end point yields the point associated with the target parameter.

The GMI is graphically demonstrated in Fig. 1. Here, the points (subspaces) $\{\mathcal{Y}_j\}_{j=1}^{N_p}$ on the Grassmann manifold $\mathcal{G}$ are spanned by the DMD modes $\{\Phi_j\}_{j=1}^{N_p}$. A reference point $\mathcal{Y}_0$ is selected from the set

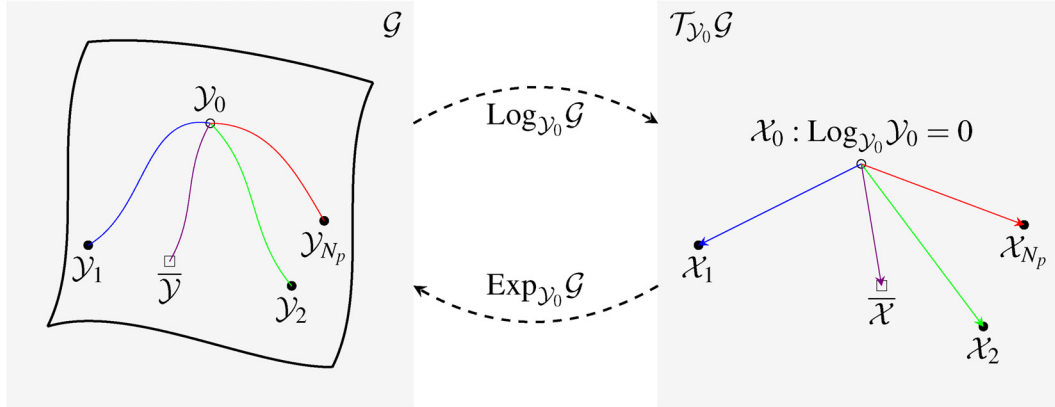


FIG. 1. Illustration of GMI.

$\{\mathcal{Y}_j\}_{j=1}^{N_p}$ to serve as the origin of the tangent space $\mathcal{T}_{\mathcal{Y}_0}\mathcal{G}$ of the Grassmann manifold. The points $\{\mathcal{Y}_j\}_{j=1}^{N_p}$ (including $\mathcal{Y}_0$ identical to some $\mathcal{Y}_j$) are then projected onto this tangent space. Thereafter, an initial derivative $\bar{\mathcal{X}}$, associated with the target parameter $\bar{\gamma}$ and represented by the matrix $\bar{\Gamma}$, can be constructed by interpolating the projections $\{\mathcal{X}_j\}_{j=1}^{N_p}$ of the points $\{\mathcal{Y}_j\}_{j=1}^{N_p}$. The subspace $\bar{\mathcal{Y}}$ on the manifold, spanned by the matrix $\bar{\Phi}$ (i.e., the DMD modes corresponding to $\bar{\gamma}$), can then be recovered from $\bar{\Gamma}$. The interpolation procedure consists of the following operations:

These points $\{\mathcal{Y}_j\}_{j=1}^{N_p}$, which are sufficiently close to the reference point $\mathcal{Y}_0$ (as determined by the distance metric), are first mapped to the tangent space $\mathcal{T}_{\mathcal{Y}_0}\mathcal{G}$. The projection of $\mathcal{Y}_j$, that is, $\mathcal{X}_j \in \mathcal{T}_{\mathcal{Y}_0}\mathcal{G}$, is represented by the matrix Γ_j obtained via the logarithmic mapping $\text{Log}_{\mathcal{Y}_0}\mathcal{G} : \mathcal{G} \mapsto \mathcal{T}_{\mathcal{Y}_0}\mathcal{G}$,

$$(I - \Phi_0(\Phi_0^*\Phi_0)^{-1}\Phi_0^*)\Phi_j(\Phi_0^*\Phi_j)^{-1}(\Phi_0^*\Phi_0)^{1/2} = M_j\hat{\Xi}_jN_j^* \quad (\text{thin SVD}),$$

$$\Gamma_j = M_j \tan^{-1} \hat{\Xi}_j N_j^*. \quad (12)$$

It is worth noting that $\text{Log}_{\mathcal{Y}_0}\mathcal{Y}_0 = 0$ acts as the origin of the tangent space $\mathcal{T}_{\mathcal{Y}_0}\mathcal{G}$. Since this tangent space is a vector space, interpolation can be performed directly as in Euclidean space. The matrix $\bar{\Gamma}$ associated with $\bar{\gamma}$, is computed as

$$\bar{\Gamma} = \sum_{j=1}^{N_p} \sigma_j \Gamma_j, \quad (13)$$

where σ_j are interpolation weights constructed using Lagrange polynomials. Ultimately, $\bar{\Gamma}$ is mapped back onto the manifold $\mathcal{G}$ to obtain $\bar{\mathcal{Y}} \in \mathcal{G}$, spanned by $\bar{\Phi}$. The exponential mapping $\text{Exp}_{\mathcal{Y}_0}\mathcal{G} : \mathcal{T}_{\mathcal{Y}_0}\mathcal{G} \mapsto \mathcal{G}$ is given by

$$\bar{\Gamma} = \bar{M} \bar{\Xi} \bar{N}^* \quad (\text{thin SVD}), \quad (14)$$

$$\bar{\Phi} = (\Phi_0(\Phi_0^*\Phi_0)^{-1/2}\bar{N} \cos \bar{\Xi} \bar{N}^* + \bar{M} \sin \bar{\Xi} \bar{N}^*)(\Phi_0^*\Phi_0)^{1/2},$$

where $\bar{\Phi}$ denotes the DMD modes corresponding to $\bar{\gamma}$.

2. Mode-realigned pointwise interpolation

MRPWI provides a numerically accurate, computationally efficient, and theoretically straightforward approach for interpolation.²⁶ This method involves realigning the DMD modes associated with different parameters to the reference DMD modes, followed by interpolation to construct the DMD modes corresponding to the target parameter from these aligned modes.

As previously mentioned, DMD modes are the columns of $\Phi_j = [\phi_j^1, \phi_j^2, \dots, \phi_j^{N_r}]$, where $\Phi_j \in \mathbb{C}^{N_n \times N_r}$ and $\phi_j^k \in \mathbb{C}^{N_n}$ for $k = 1, 2, \dots, N_r$. Initially, reference modes Φ_0 are selected from the set $\{\Phi_j\}_{j=1}^{N_p}$. The modes $\{\Phi_j\}_{j=1}^{N_p}$ are then rotated column-wise to align with the reference modes $\Phi_0 = [\phi_0^1, \phi_0^2, \dots, \phi_0^{N_r}]$, as follows:

$$\phi_j^k \rightarrow \phi_j^k e^{-i\varphi_j^k}, \quad \text{for } j = 1, \dots, N_p, \quad k = 1, 2, \dots, N_r, \quad (15)$$

where i is the imaginary unit, and $\varphi_j^k = \varphi(\phi_0^k, \phi_j^k)$ is the (Kasner's) pseudo-angle. Specifically, the pseudo-angle φ between the complex vectors ϕ_a and ϕ_b is defined as

$$e^{i\varphi(\phi_a, \phi_b)} = \frac{\langle \phi_a, \phi_b \rangle}{\|\phi_a\| \|\phi_b\|} \Rightarrow \varphi(\phi_a, \phi_b) = -i \ln \frac{\langle \phi_a, \phi_b \rangle}{\|\phi_a\| \|\phi_b\|}.$$

Here, $\varphi \in [-\pi, \pi]$, and the notation $\langle \phi_a, \phi_b \rangle$ denotes the complex inner product $\phi_a^* \phi_b$. Subsequently, the DMD modes $\bar{\Phi}$ at the target parameter $\bar{\gamma}$ are computed from the aligned DMD modes $\{\Phi_j\}_{j=1}^{N_p}$ in combination with Lagrange interpolation,

$$\bar{\Phi} = \sum_{j=1}^{N_p} \sigma_j \Phi_j, \quad (16)$$

where σ_j represents the interpolation weights.

III. CASE STUDY: FLOW OVER A CYLINDER

A. Flow description and numerical framework

The governing equations for unsteady, two-dimensional, incompressible fluid flows are given by the non-dimensional Navier–Stokes equations,

$$\nabla \cdot \mathbf{u} = 0, \quad \partial \mathbf{u} / \partial t + \mathbf{u} \cdot \nabla \mathbf{u} = -\nabla p + Re^{-1} \nabla^2 \mathbf{u}, \quad (17)$$

where the velocity $\mathbf{u}(\mathbf{x}, t) = (u, v)^T$ and pressure $p(\mathbf{x}, t)$ are defined over spatial coordinates $\mathbf{x} = (x, y)^T$ and time t . The Reynolds number, defined as $Re = U_\infty D / \nu$, is determined by the freestream velocity U_∞ , cylinder diameter D , and kinematic viscosity ν .

Figure 2 illustrates the spatial domain setup for flow over a cylinder. Specifically, a Cartesian coordinate system is established with the streamwise direction along x , the transverse direction along y , and the origin located at the center of the cylinder. The rectangular domain measures $50D \times 30D$ in the streamwise and transverse directions, respectively. At the inlet $x = -15D$, a Dirichlet boundary condition is applied, specifying $\mathbf{u}(\mathbf{x}, t) = (U_\infty, 0)^T$ with $U_\infty = 1$ uniformly.³⁵ Meanwhile, a natural condition, formulated as $[-p\mathbf{I} + \nu \nabla \mathbf{u}] \cdot \mathbf{n} = 0$, is imposed at the outlet $x = 35D$, where $\mathbf{n}$ denotes the unit normal vector.³⁶ The cylinder surface satisfies the no-slip condition, ensuring zero velocity relative to the wall. The top and bottom boundaries ($y = \pm 15D$) are assigned periodic boundary conditions to simulate an unbounded flow domain and minimize wall effects.³⁷

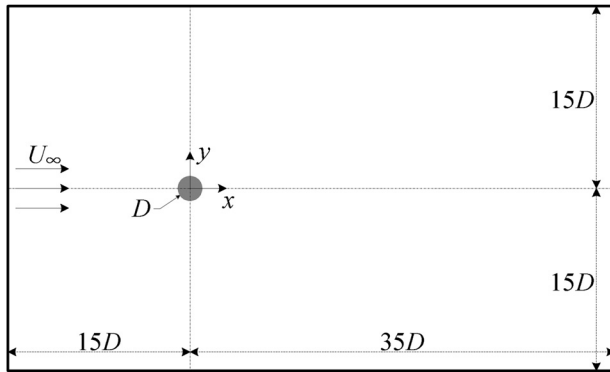


FIG. 2. Spatial domain configuration for flow over a cylinder, with the cylinder diameter doubled for clarity.

The computational domain is initialized with the inlet velocity profile, and periodic vortex shedding develops in the flow field after sufficient computational time.

Direct numerical simulations (DNS) are conducted using the open-source spectral element code Nek5000.^{38,39} A total of 1472 spectral elements constitute the computational domain, with $8^2 = 64$ Gauss-Lobatto-Legendre (GLL) quadrature points in each element. Specifically, seventh-order Lagrange polynomials are defined along each direction within a spectral element to represent velocity and pressure, corresponding to eight GLL points and six Gauss-Legendre (GL) points, respectively ($\mathbb{P}_N - \mathbb{P}_{N-2}$ formulation).³⁶ The temporal discretization employs a third-order backward difference formula/extrapolation (BDF3/EXT3), combining implicit treatment of viscous terms (BDF3) and explicit integration of convective terms (EXT3),⁴⁰ with a fixed time step $\Delta\tau = 0.001$.

A total of 19 flow cases are simulated, parameterized by Reynolds numbers spanning $Re \in \{70, 80, 85, 90, 100, 110, 115, 120, 125, 130, 135, 140, 145, 150, 160, 170, 175, 180, 190\}$. These flows exhibit two-dimensional laminar dynamics with periodic vortex shedding ($47 < Re < 200$).⁴¹ Flow snapshots are collected during the periodic vortex shedding regime at intervals of $\Delta t = 100\Delta\tau = 0.1$, sampling three periods of vortex shedding. This ensures compliance with the Nyquist-Shannon criterion, which requires the sampling frequency to be at least twice the inherent frequency (i.e., the dominant vortex shedding frequency).^{9,34} These flow data spanning multiple Reynolds numbers serve two purposes: (1) the construction of DMD-Galerkin PROMs, and (2) the subsequent evaluation of these PROMs at target Reynolds numbers $\bar{Re}$. The sensitivity of PROMs developed using the GMI and MRPWI methods is first comprehensively investigated by varying the number of modes N_r , the system parameter interval ΔRe , and the number of neighbors N_p , with $\bar{Re} = 130$ serving as the benchmark configuration. Then, these PROMs are examined under specific parameter settings, with an emphasis on analyzing flow fields and aerodynamic coefficients. Finally, the generalization capability of these PROMs is systematically evaluated through both interpolation within the parameter space and extrapolation to unobserved regimes.

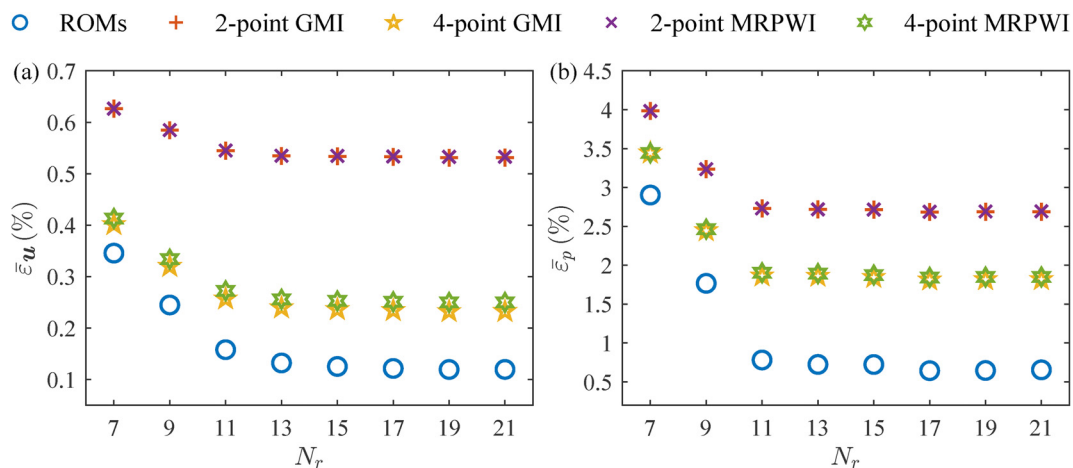


FIG. 3. Error trends (RLEs) vs number of modes N_r . (a) Velocity and (b) pressure (2-point: $Re \in \{120, 140\}$; 4-point: $Re \in \{100, 120, 140, 160\}$).

B. Demonstration of DMD-Galerkin parametric reduced-order models

After constructing the DMD-Galerkin PROMs for target parameters, their performance is assessed through relative L^2 errors (RLEs), where the relative L^2 error metric is defined as

$$\bar{\varepsilon} = \sqrt{\frac{\sum_{j=1}^{N_r} \|s_j^{\text{ROM}} - s_j^{\text{DNS}}\|_2^2}{\sum_{j=1}^{N_r} \|s_j^{\text{DNS}}\|_2^2}}. \quad (18)$$

In this expression, s_j denotes the j th snapshot from DNS or a (non)-parametric reduced-order model (ROM), with DNS acting as the benchmark, and the average is taken over N_r snapshots.

For the target parameter $\overline{Re} = 130$, the DMD-Galerkin PROMs are developed through GMI or MRPWI under various settings of N_r , ΔRe , and N_p , representing the number of modes, system parameter interval, and number of neighbors, respectively. As shown in Fig. 3, the RLEs for velocity and pressure decrease with increasing N_r and stabilize once certain values of N_r are reached, with the 2-point and 4-point interpolation schemes ($N_p \in \{2, 4\}$) and $\Delta Re = 20$. Figure 4 illustrates that the RLEs increase with expanding system parameter interval ΔRe , as observed for both the 2-point and 4-point interpolation methods ($N_p \in \{2, 4\}$) with $N_r = 13$. Increasing the number of neighbors N_p in higher-order interpolation reduces the RLEs, as demonstrated in Fig. 5, with $N_r = 13$ and $\Delta Re = 20$.

It is observed that the predictive accuracy of PROMs improves with an increasing number of modes N_r and neighbors N_p , whereas a smaller parameter interval ΔRe further enhances their fidelity. The PROMs constructed by GMI and MRPWI maintain comparable

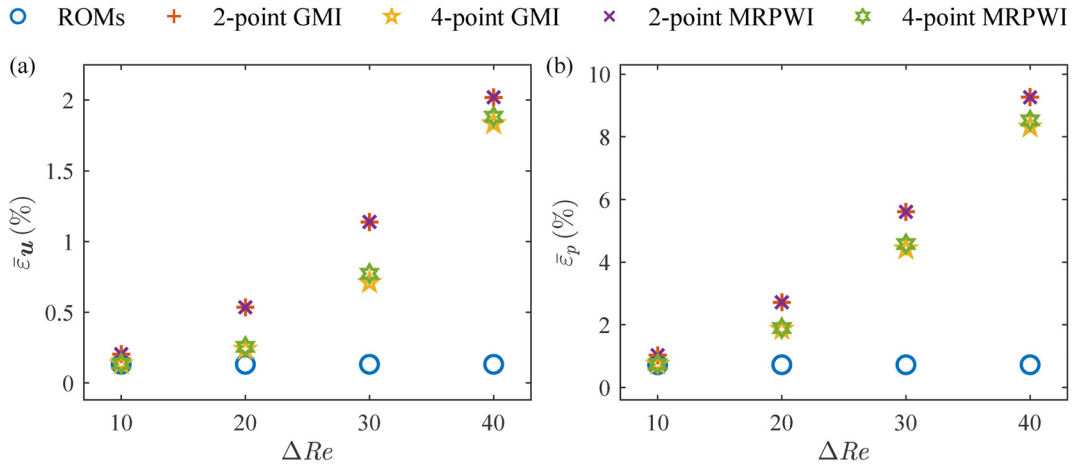


FIG. 4. Error trends (RLEs) vs system parameter interval ΔRe . (a) Velocity and (b) pressure (2-point: $Re \in \{125, 135\}, \{120, 140\}, \{115, 145\}, \{110, 150\}$; 4-point: $Re \in \{115, 125, 135, 145\}, \{100, 120, 140, 160\}, \{85, 115, 145, 175\}, \{70, 110, 150, 190\}$).

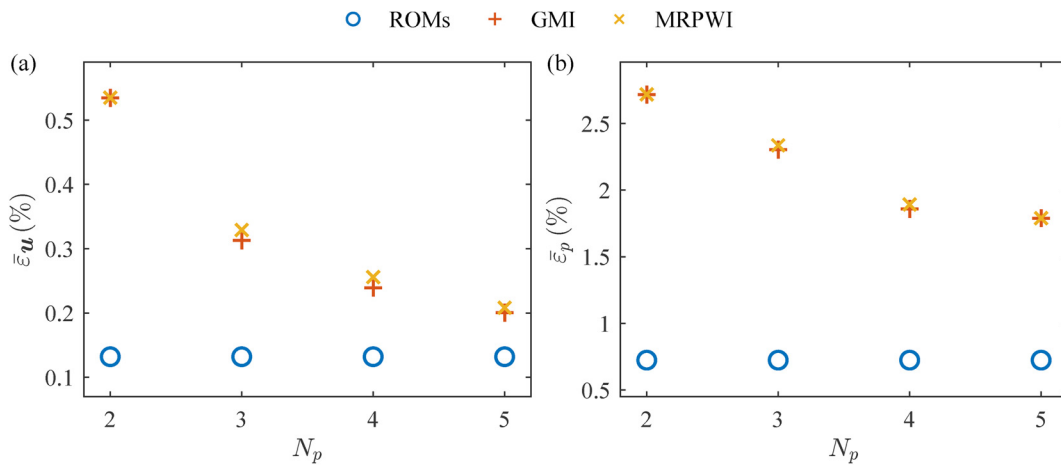


FIG. 5. Error trends (RLEs) vs number of neighbors N_p . (a) Velocity and (b) pressure ($Re \in \{120, 140\}, \{100, 120, 140\}, \{100, 120, 140, 160\}, \{80, 100, 120, 140, 160\}$).

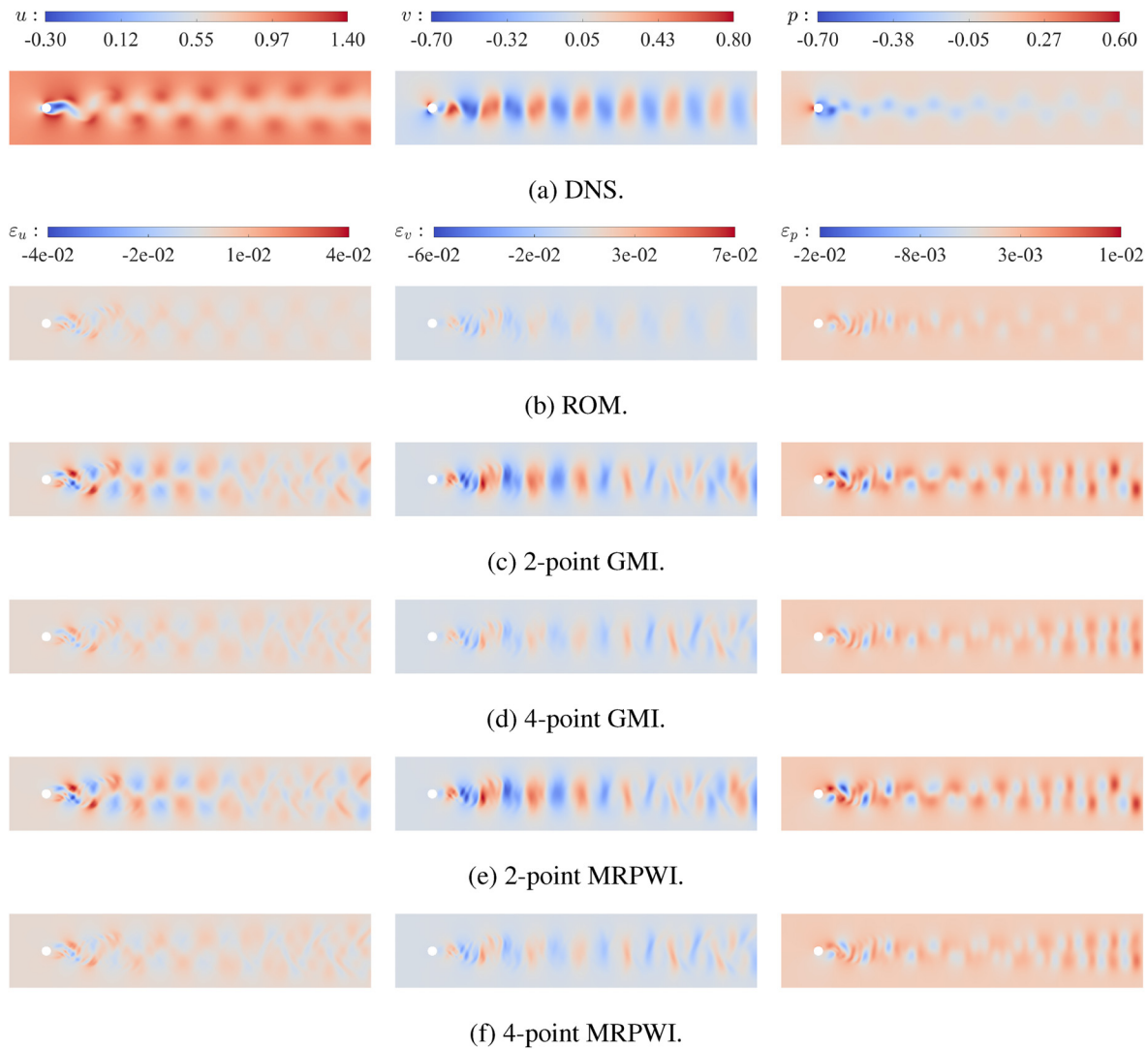


FIG. 6. DNS flow fields and ROM/PROMs error distributions at $t = 15$.

accuracy across different settings of N_r , ΔRe , and N_p . Based on this, intermediate settings with $N_r = 13$, $\Delta Re = 20$, and $N_p \in \{2, 4\}$, corresponding to $Re \in \{120, 140\}$ and $\{100, 120, 140, 160\}$, respectively, are selected for subsequent analysis and demonstration.

At $\overline{Re} = 130$, the DMD-Galerkin PROMs exhibit high accuracy, with the evolution of flow fields and aerodynamic coefficients closely aligning with those obtained from DNS and the DMD-Galerkin ROM. The RLEs are no more than 0.6% for velocity and 3.0% for pressure. Figure 6 better demonstrates the distributions of flow fields and errors of the ROM and PROMs relative to DNS. The first row depicts the flow fields from DNS [Fig. 6(a)], while the subsequent rows [Figs. 6(b)–6(f)] show the error distributions $\varepsilon_j = \mathbf{s}_j^{\text{ROM}} - \mathbf{s}_j^{\text{DNS}}$ from the ROM and PROMs, with normalization using corresponding scales. These fields reflect that velocity components $\mathbf{u} = (u, v)^T$ and pressure p predicted by the PROMs at $t = 15$ exhibit high accuracy. The error patterns and magnitudes of GMI and MRPWI are in close agreement

at identical N_p . Clearly, the errors corresponding to 4-point interpolation methods are significantly lower than those generated by 2-point methods, and are even comparable to those of the ROM. The consistency between high-amplitude regions in DNS fields and error fields indicates that one of the dominant parts of error is amplitude discrepancy. In addition, there are oscillations at smaller scales in regions very close to and far from the cylinder.

The lift and drag coefficients are computed from the pressure and viscous forces exerted on the cylinder surface, which are determined by surface integration over the boundary of the cylinder. The dimensionless coefficients and the force are given by

$$C_D = \frac{F_x}{\rho U_\infty^2 D/2}, \quad C_L = \frac{F_y}{\rho U_\infty^2 D/2}, \quad \mathbf{F} = \oint_{\partial\Omega} (-p\mathbf{I} + \boldsymbol{\tau}) \cdot \mathbf{n} dS. \quad (19)$$

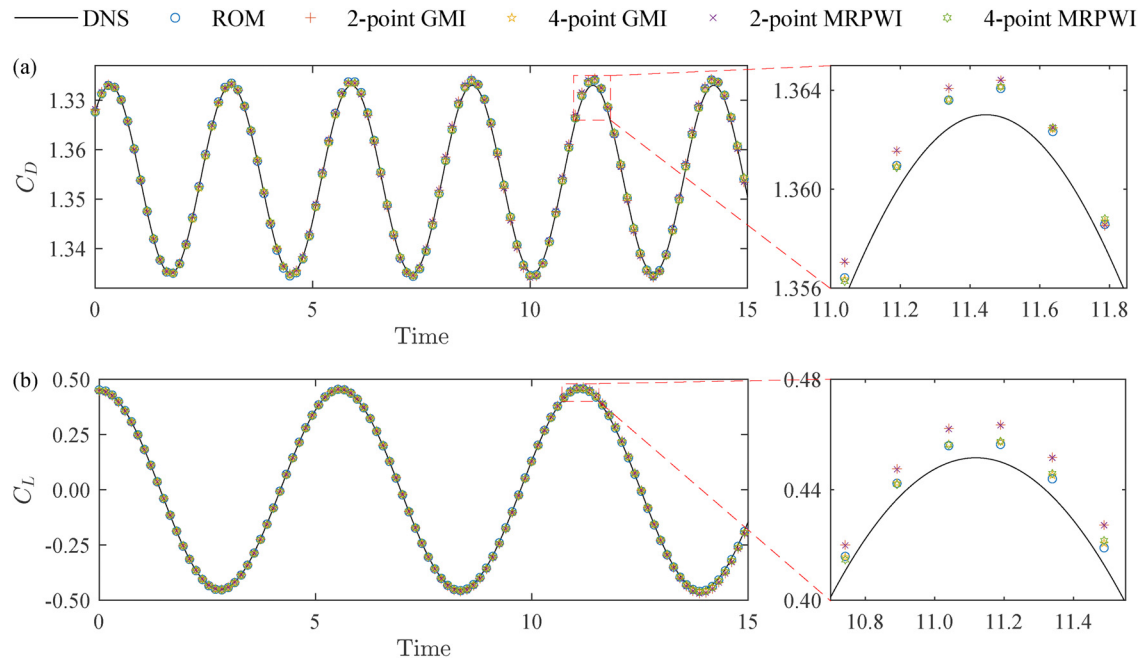


FIG. 7. Time evolution of drag and lift coefficients, with zoomed-in views on the right. (a) Drag and (b) lift.

The force vector $\mathbf{F} = (F_x, F_y)$ is determined by the pressure p , viscous stress tensor $\boldsymbol{\tau}$, and the outward normal vector $\mathbf{n}$ evaluated along the cylinder surface $\partial\Omega$, where ρ is the fluid density, U_∞ is the freestream velocity, and D is the diameter of the cylinder. Figure 7 illustrates the time evolution of the drag and lift coefficients, demonstrating the high fidelity of PROMs constructed with various interpolation methods in comparison to DNS and the ROM. Specifically, the accuracy of these PROMs is more clearly visible in the zoomed-in views, further verifying the preceding analysis: GMI- and MRPWI-generated PROMs exhibit comparable accuracy and progressively enhanced performance with increasing N_p .

The generalization capability of these PROMs is assessed by applying the intermediate configuration consisting of $Re \in \{120, 140\}$ and $\{100, 120, 140, 160\}$ with $N_r = 13$, $\Delta Re = 20$, and $N_p \in \{2, 4\}$. As shown in Fig. 8, the RLEs are evaluated across the interpolation region ($\overline{Re} \in [110, 150]$) and extrapolation ranges ($\overline{Re} \in [80, 90] \cup [170, 180]$). Within the interpolation range, the constructed PROMs achieve higher accuracy. The RLEs are below 0.8% (velocity) and 3.9% (pressure) for 2-point interpolation, and 0.5% (velocity) and 3.7% (pressure) for 4-point interpolation. In contrast, the extrapolation cases exhibit significantly larger RLEs, particularly for 4-point interpolation, which is more strongly influenced by distant neighboring

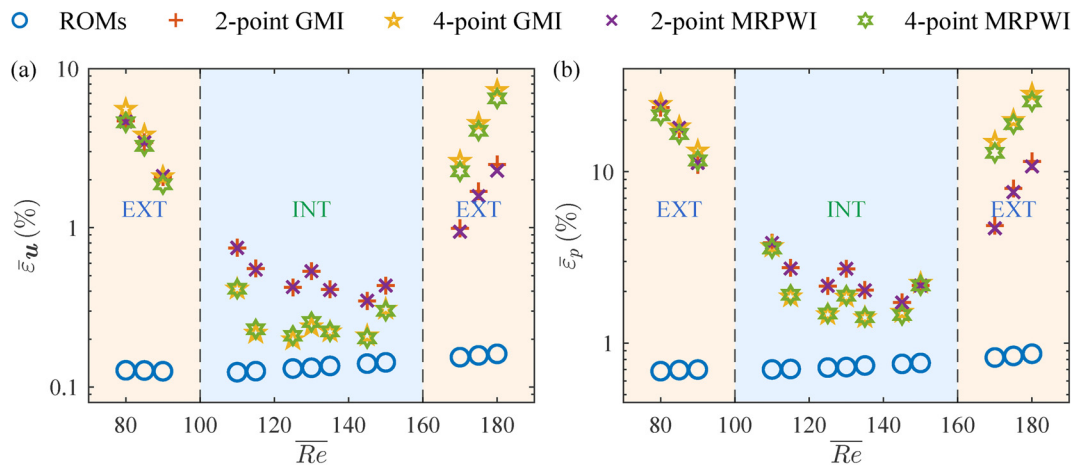


FIG. 8. Error trends (RLEs) vs target Reynolds number $\overline{Re}$. (a) Velocity and (b) pressure. INT: interpolation; EXT: extrapolation ($Re \in \{100, 120, 140, 160\}$ with $N_r = 13$, $\Delta Re = 20$ and $N_p \in \{2, 4\}$).

points. Consequently, 4-point extrapolation may exhibit poorer performance than 2-point extrapolation when $\overline{Re}$ exceeds the upper bound of the interpolation range. These results demonstrate that while the PROMs maintain high accuracy in the interpolation range, their predictive capability deteriorates significantly in extrapolation regimes.

IV. CONCLUSION

The DMD-Galerkin PROMs are constructed by combining DMD-Galerkin framework with various interpolation methods. Specifically, the complex-valued, non-orthonormal DMD modes at the target parameter are computed with either GMI or MRPWI, which are then coupled with the Galerkin projection to form the DMD-Galerkin PROMs. These PROMs are assessed and demonstrated in the case of flow over a cylinder, where they exhibit significant accuracy in comparison to DNS and DMD-Galerkin ROMs. Across different settings, including the number of modes N_r , system parameter interval ΔRe , and number of neighbors N_p , the PROMs constructed with GMI and MRPWI exhibit comparable accuracy. With increasing N_r , increasing N_p , or decreasing ΔRe in the interpolation, the accuracy of these PROMs is further enhanced. In practice, such monotonic patterns may not be universal, and the optimal settings for N_r , $\Delta \gamma$, and N_p should be determined empirically or based on available data in general. While the PROMs demonstrate high-fidelity accuracy within the investigated interpolation range, their extrapolation performance degrades substantially, exhibiting significantly larger error magnitudes. Generally, it is better that target parameters lie within the interpolation range of the parameter space, preferably near its center, while extrapolation should be restricted to regions close to the parameter space boundaries for accurate PROMs. High-order schemes are recommended for interpolation, while lower-order schemes generally provide more reliable results for extrapolation. Additionally, the PROMs constructed with MRPWI are more efficient than those built with GMI, attributable to its theoretical simplicity.

The DMD-Galerkin PROMs, along with the GMI and MRPWI methods, are well-suited for extension to reduced-order modeling in other areas owing to their accuracy and efficiency.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Lei Du: Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Software (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Shengqi Zhang:** Conceptualization (equal); Formal analysis (equal); Funding acquisition (equal); Methodology (equal); Project administration (equal);

Resources (equal); Supervision (equal); Writing – review & editing (equal). **Yongchao Yang:** Funding acquisition (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

APPENDIX: DERIVATIONS OF EXPONENTIAL AND LOGARITHMIC MAPPINGS ON THE COMPLEX GRASSMANN MANIFOLD VIA THE NONCOMPACT STIEFEL MANIFOLD REPRESENTATION

Following the derivation of the geodesic, exponential, and logarithmic mappings on the real Grassmann manifold,^{42,43} as well as on the complex Grassmann manifold with a Stiefel manifold representation,²⁶ the corresponding counterparts based on the noncompact Stiefel manifold representation are accordingly established.

The set of all p -dimensional subspaces of the Euclidean space $\mathbb{C}^n$, where $p \leq n$, is known as the Grassmann manifold (also called the Grassmannian), denoted by $\mathcal{G} = Gr(p, n)$,

$$Gr(p, n) := \{\mathcal{Y} \subset \mathbb{C}^n | \mathcal{Y} \text{ is a subspace, } \dim(\mathcal{Y}) = p\}. \quad (\text{A1})$$

It can be viewed as a Riemannian quotient manifold of the noncompact Stiefel manifold $St(p, n)$,

$$Gr(p, n) = St(p, n)/GL(p) = \{[\Psi] | \Psi \in St(p, n)\}, \quad (\text{A2})$$

where the noncompact Stiefel manifold $St(p, n)$ is the set of all $n \times p$ complex matrices with full column rank, defined as

$$St(p, n) := \{\Psi \in \mathbb{C}^{n \times p} | \text{rank}(\Psi) = p\}. \quad (\text{A3})$$

Each point (subspace) $\mathcal{Y} \in \mathcal{G}$ is identified with an equivalence class $[\Psi]$, where $[\Psi] = \{\Psi R | R \in GL(p)\} = \Psi GL(p)$. That is, $\mathcal{Y} = [\Psi]$ represents all matrices related to $\Psi \in \mathbb{C}^{n \times p}$ by right multiplication with an invertible $p \times p$ matrix $GL(p)$. Equivalently, the subspace $\mathcal{Y}$ is given by the column space of Ψ , that is, $\mathcal{Y} = \text{span}(\Psi)$. The matrix representation of a subspace is thus not unique; any matrix whose columns span the same subspace belongs to the same equivalence class. At any point $\mathcal{Y} = [\Psi] \in \mathcal{G}$, the tangent space $\mathcal{T}_{\mathcal{Y}}\mathcal{G}$ is given by

$$\mathcal{T}_{\mathcal{Y}}\mathcal{G} = \{\Gamma \in \mathbb{C}^{n \times p} | \Psi^* \Gamma = 0\}, \quad (\text{A4})$$

with $\mathcal{Y}$ serving as the origin of the tangent space. The tangent space $\mathcal{T}_{\mathcal{Y}}\mathcal{G}$ is a flat vector space with the same dimension as the manifold $\mathcal{G}$, allowing interpolation to be carried out in $\mathcal{T}_{\mathcal{Y}}\mathcal{G}$ in a manner analogous to that in Euclidean space. The Riemannian metric, an inner product on the tangent space $\mathcal{T}_{\mathcal{Y}}\mathcal{G}$, is admitted as

$$\langle \Gamma_1, \Gamma_2 \rangle_{\mathcal{Y}} = \text{Re} \left[\text{tr} \left((\Psi^* \Psi)^{-1} \Gamma_1^* \Gamma_2 \right) \right], \quad \Gamma_1, \Gamma_2 \in \mathcal{T}_{\mathcal{Y}}\mathcal{G}. \quad (\text{A5})$$

To derive the geodesics equation on the Grassmann manifold $\mathcal{G}$, consider the minimization of the energy functional,

$$\begin{aligned} \mathcal{E}[\Psi(t), \Psi^*(t)] &= \int_0^1 \frac{1}{2} \text{tr}((\Psi^* \Psi)^{-1} \dot{\Psi}^* \dot{\Psi}) dt \\ &= \int_0^1 \mathcal{F}(t, \Psi, \dot{\Psi}, \Psi^*, \dot{\Psi}^*) dt. \end{aligned} \quad (\text{A6})$$

Since both $(\Psi^* \Psi)^{-1}$ and $\dot{\Psi}^* \dot{\Psi}$ are Hermitian matrices, the trace of their product is real-valued, so the use of the $\text{Re}[\cdot]$ operator is unnecessary.⁴⁴ By applying the variation $\Psi(t) \rightarrow \Psi(t) + \delta\Psi(t)$ and its conjugate $\Psi^*(t) \rightarrow \Psi^*(t) + \delta\Psi^*(t)$, the first variation of the action becomes

$$\delta\mathcal{E} = \int_0^1 \left[\left(\frac{\partial\mathcal{F}}{\partial\Psi} - \frac{d}{dt} \frac{\partial\mathcal{F}}{\partial\dot{\Psi}} \right) \delta\Psi + \left(\frac{\partial\mathcal{F}}{\partial\Psi^*} - \frac{d}{dt} \frac{\partial\mathcal{F}}{\partial\dot{\Psi}^*} \right) \delta\Psi^* \right] dt. \quad (\text{A7})$$

The Euler–Lagrange equations follow from the stationarity condition $\delta\mathcal{E} = 0$, leading to the geodesic equation,

$$\ddot{\Psi}(t) + \Psi(t)(\Psi^* \Psi)^{-1} \dot{\Psi}^* \dot{\Psi} = 0, \quad t \in [0, 1]. \quad (\text{A8})$$

Since $\dot{\Psi}$ lies in the tangent space $\mathcal{T}_{\mathcal{Y}}\mathcal{G}$, it satisfies the orthogonality condition $\dot{\Psi}^* \dot{\Psi} = 0$ according to the definition in Eq. (A4). Differentiating $\dot{\Psi}^* \dot{\Psi}$ with respect to the parameter t , and employing the orthogonality condition $\dot{\Psi}^* \Psi = (\Psi^* \dot{\Psi})^* = 0$, together with the geodesic equation [Eq. (A8)], yields

$$\begin{aligned} \frac{d}{dt} \dot{\Psi}^* \dot{\Psi} &= \dot{\Psi}^* \ddot{\Psi} + \ddot{\Psi}^* \dot{\Psi} \\ &= (-\Psi(\Psi^* \Psi)^{-1} \dot{\Psi}^* \dot{\Psi})^* \dot{\Psi} - \dot{\Psi}^* \Psi(\Psi^* \Psi)^{-1} \dot{\Psi}^* \dot{\Psi} = 0, \end{aligned} \quad (\text{A9})$$

which indicates that $\dot{\Psi}^* \dot{\Psi}$ remains constant over t . Let $\mathcal{Y}(0) = [\Psi_0]$, and given that $\dot{\Psi}_0(\Psi^* \Psi)^{-1/2} = M\Xi N^*$, the relation $(\Psi^* \Psi)^{-1/2} \dot{\Psi}^* \dot{\Psi} (\Psi^* \Psi)^{-1/2} = N\Xi^2 N^*$ follows. Inserting these expressions into Eq. (A8) leads to the equation,

$$\ddot{\Psi}(t)(\Psi^* \Psi)^{-1/2} N + \Psi(t)(\Psi^* \Psi)^{-1/2} N\Xi^2 = 0. \quad (\text{A10})$$

The solution to the above second-order differential equation is

$$\Psi(t)(\Psi_0^* \Psi_0)^{-1/2} N = \Psi_0(\Psi_0^* \Psi_0)^{-1/2} N \cos(\Xi t) + M \sin(\Xi t). \quad (\text{A11})$$

Hence, the geodesic path is described by

$$\begin{aligned} \mathcal{Y}(t) = \text{span}(\Psi(t)) &= \text{span} \left((\Psi_0(\Psi_0^* \Psi_0)^{-1/2} N \cos(\Xi t) N^* \right. \\ &\quad \left. + M \sin(\Xi t) N^*) (\Psi_0^* \Psi_0)^{1/2} \right). \end{aligned} \quad (\text{A12})$$

Ultimately, the exponential mapping, which maps a tangent vector from the tangent space to the end point of the corresponding geodesic on the Grassmann manifold, is defined as $\text{Exp}_{\mathcal{Y}_0} \mathcal{G} : \mathcal{T}_{\mathcal{Y}_0} \mathcal{G} \ni \Gamma \rightarrow \Psi \in \mathcal{G}$,

$$\begin{aligned} \Gamma &= M\Xi N^* \quad (\text{thin SVD}), \\ \Psi &= (\Psi_0(\Psi_0^* \Psi_0)^{-1/2} N \cos \Xi N^* + M \sin \Xi N^*) (\Psi_0^* \Psi_0)^{1/2}. \end{aligned} \quad (\text{A13})$$

The logarithmic mapping, that is, the inverse of the exponential mapping, projects a point on the Grassmann manifold onto the tangent space. It is given by $\text{Log}_{\mathcal{Y}_0} \mathcal{G} : \mathcal{G} \ni \Psi \rightarrow \Gamma \in \mathcal{T}_{\mathcal{Y}_0} \mathcal{G}$,

$$\begin{aligned} (I - \Psi_0(\Psi_0^* \Psi_0)^{-1} \Psi_0^*) \Psi (\Psi_0^* \Psi)^{-1} (\Psi_0^* \Psi_0)^{1/2} &= M\hat{\Xi} N^* \quad (\text{thin SVD}), \\ \Gamma &= M \tan^{-1} \hat{\Xi} N^*. \end{aligned} \quad (\text{A14})$$

It should be noted that the equality holds by combining Eqs. (A13) and (A14), leading to

$$(I - \Psi_0(\Psi_0^* \Psi_0)^{-1} \Psi_0^*) \Psi (\Psi_0^* \Psi)^{-1} (\Psi_0^* \Psi_0)^{1/2} = M \tan \Xi N^*. \quad (\text{A15})$$

Thus,

$$\begin{aligned} (I - \Psi_0(\Psi_0^* \Psi_0)^{-1} \Psi_0^*) \Psi (\Psi_0^* \Psi)^{-1} (\Psi_0^* \Psi_0)^{1/2} \\ = M\hat{\Xi} N^* \Rightarrow \Gamma = M \tan^{-1} \hat{\Xi} N^*. \end{aligned} \quad (\text{A16})$$

REFERENCES

- P. Benner, S. Gugercin, and K. Willcox, "A survey of projection-based model reduction methods for parametric dynamical systems," *SIAM Rev.* **57**, 483–531 (2015).
- J. Kou and W. Zhang, "Data-driven modeling for unsteady aerodynamics and aeroelasticity," *Prog. Aerosp. Sci.* **125**, 100725 (2021).
- P. Benner, S. Grivet-Talocia, A. Quarteroni, G. Rozza, W. Schilders, and L. M. Silveira, *System and Data-Driven Methods and Algorithms* (De Gruyter, 2021).
- S. L. Brunton, B. R. Noack, and P. Koumoutsakos, "Machine learning for fluid mechanics," *Annu. Rev. Fluid Mech.* **52**, 477–508 (2020).
- K. Hasegawa, K. Fukami, T. Murata, and K. Fukagata, "CNN-LSTM based reduced order modeling of two-dimensional unsteady flows around a circular cylinder at different Reynolds numbers," *Fluid Dyn. Res.* **52**, 065501 (2020).
- A. Prakash and Y. J. Zhang, "Nonintrusive projection-based reduced order modeling using stable learned differential operators," *Comput. Methods Appl. Mech. Eng.* **442**, 117946 (2025).
- J. L. Lumley, *Stochastic Tools in Turbulence* (Academic Press, 1970).
- L. Sirovich, "Turbulence and the dynamics of coherent structures. I. Coherent structures," *Q. Appl. Math.* **45**, 561–571 (1987).
- P. J. Schmid, "Dynamic mode decomposition of numerical and experimental data," *J. Fluid Mech.* **656**, 5–28 (2010).
- C. W. Rowley and S. T. Dawson, "Model reduction for flow analysis and control," *Annu. Rev. Fluid Mech.* **49**, 387–417 (2017).
- K. Taira, M. S. Hemati, S. L. Brunton, Y. Sun, K. Duraisamy, S. Bagheri, S. T. Dawson, and C.-A. Yeh, "Modal analysis of fluid flows: Applications and outlook," *AIAA J.* **58**, 998–1022 (2020).
- P. J. Schmid and J. Sesterhenn, "Dynamic mode decomposition of numerical and experimental data," in *61st Annual Meeting of the APS Division of Fluid Dynamics* (American Physical Society, 2008), p. 101, see <https://meetings.aps.org/Meeting/DFD08/Event/91003>.
- C. W. Rowley, I. Mezić, S. Bagheri, P. Schlatter, and D. S. Henningson, "Spectral analysis of nonlinear flows," *J. Fluid Mech.* **641**, 115–127 (2009).
- J. N. Kutz, S. L. Brunton, B. W. Brunton, and J. L. Proctor, *Dynamic Mode Decomposition: Data-Driven Modeling of Complex Systems* (SIAM, 2016).
- S. L. Brunton and J. N. Kutz, *Data-Driven Science and Engineering: Machine Learning, Dynamical Systems, and Control* (Cambridge University Press, 2022).
- T. Sayadi, P. J. Schmid, F. Richecoeur, and D. Durox, "Parametrized data-driven decomposition for bifurcation analysis, with application to thermo-acoustically unstable systems," *Phys. Fluids* **27**, 037102 (2015).
- Q. A. Huhn, M. E. Tano, J. C. Ragusa, and Y. Choi, "Parametric dynamic mode decomposition for reduced order modeling," *J. Comput. Phys.* **475**, 111852 (2023).
- M. W. Hess, A. Quaini, and G. Rozza, "A data-driven surrogate modeling approach for time-dependent incompressible Navier-Stokes equations with dynamic mode decomposition and manifold interpolation," *Adv. Comput. Math.* **49**, 22 (2023).
- D. Amsallem, *Interpolation on Manifolds of CFD-Based Fluid and Finite Element-Based Structural Reduced-Order Models for on-Line Aeroelastic Predictions* (Stanford University, 2010).
- K. Li, Y. Li, L. Li, and S. Lanteri, "Surrogate modeling of time-domain electromagnetic wave propagation via dynamic mode decomposition and radial basis function," *J. Comput. Phys.* **491**, 112354 (2023).
- Z. Gao, Y. Lin, X. Sun, and X. Zeng, "A reduced order method for nonlinear parameterized partial differential equations using dynamic mode decomposition coupled with k-nearest-neighbors regression," *J. Comput. Phys.* **452**, 110907 (2022).
- H. Song, Y. Ba, D. Chen, and Q. Li, "A model reduction method for parametric dynamical systems defined on complex geometries," *J. Comput. Phys.* **506**, 112923 (2024).

- ²³G. Tissot, L. Cordier, N. Benard, and B. R. Noack, "Model reduction using dynamic mode decomposition," *C. R. Méc.* **342**, 410–416 (2014).
- ²⁴J. L. Callahan, S. L. Brunton, and J.-C. Loiseau, "On the role of nonlinear correlations in reduced-order modelling," *J. Fluid Mech.* **938**, A1 (2022).
- ²⁵W. Zhang and M. Wei, "Model order reduction using DMD modes and adjoint DMD modes," AIAA Paper No. 2017-3482, 2017.
- ²⁶L. Du, S. Zhang, R. Zhang, and S. Zhang, "Interpolation-based parametric reduced-order models with dynamic mode decomposition," *SSRN 5181891* (2025).
- ²⁷D. Amsallem and C. Farhat, "Interpolation method for adapting reduced-order models and application to aeroelasticity," *AIAA J.* **46**, 1803–1813 (2008).
- ²⁸M. Oulghelou and C. Allery, "Non intrusive method for parametric model order reduction using a bi-calibrated interpolation on the Grassmann manifold," *J. Comput. Phys.* **426**, 109924 (2021).
- ²⁹P. Holmes, *Turbulence, Coherent Structures, Dynamical Systems and Symmetry* (Cambridge University Press, 2012).
- ³⁰M. Bergmann, C.-H. Bruneau, and A. Iollo, "Enablers for robust pod models," *J. Comput. Phys.* **228**, 516–538 (2009).
- ³¹S. Lorenzi, A. Cammi, L. Luzzi, and G. Rozza, "Pod-Galerkin method for finite volume approximation of Navier–Stokes and RANS equations," *Comput. Methods Appl. Mech. Eng.* **311**, 151–179 (2016).
- ³²B. R. Noack, P. Papas, and P. A. Monkewitz, "The need for a pressure-term representation in empirical Galerkin models of incompressible shear flows," *J. Fluid Mech.* **523**, 339–365 (2005).
- ³³A. Alla and J. N. Kutz, "Nonlinear model order reduction via dynamic mode decomposition," *SIAM J. Sci. Comput.* **39**, B778–B796 (2017).
- ³⁴J. H. Tu, "Dynamic mode decomposition: Theory and applications," Ph.D. thesis (Princeton University, 2013).
- ³⁵W. Jia and H. Xu, "Effect of synthetic jets actuator parameters on deep reinforcement learning-based flow control performance in a square cylinder," *Phys. Fluids* **36**, 084113 (2024).
- ³⁶D. Massaro and P. Schlatter, "Global stability of the flow past a stepped cylinder," *J. Fluid Mech.* **988**, A1 (2024).
- ³⁷R. Bourguet, "Flow-induced vibrations of a cylinder along a circular arc," *J. Fluid Mech.* **954**, A7 (2023).
- ³⁸A. T. Patera, "A spectral element method for fluid dynamics: Laminar flow in a channel expansion," *J. Comput. Phys.* **54**, 468–488 (1984).
- ³⁹P. Fischer, J. Lottes, and S. Kerkemeier, see <http://nek5000.mcs.anl.gov> for "Nek5000: Open source spectral element CFD solver" (2008).
- ⁴⁰J. L. Callahan, J.-C. Loiseau, and S. L. Brunton, "Multiscale model reduction for incompressible flows," *J. Fluid Mech.* **973**, A3 (2023).
- ⁴¹F. Caruso Lombardi, A. Bongarzone, G. Zampogna, F. Gallaire, S. Camarri, and P. Ledda, "Von Kármán vortex street past a permeable circular cylinder: Two-dimensional flow and dynamic-mode-decomposition-based secondary stability analysis," *Phys. Rev. Fluids* **8**, 083901 (2023).
- ⁴²P.-A. Absil, R. Mahony, and R. Sepulchre, "Riemannian geometry of Grassmann manifolds with a view on algorithmic computation," *Acta Appl. Math.* **80**, 199–220 (2004).
- ⁴³G. Ciaramella, M. J. Gander, and T. Vanzan, "A gentle introduction to interpolation on the Grassmann manifold," Oberwolfach *Preprint No. 2024-02* (2024).
- ⁴⁴R. A. Horn and C. R. Johnson, *Matrix Analysis* (Cambridge University Press, 2012).